Q.1 what is the total revenue generated by male vs female customer

```
select gender ,sum(purchase_amount) as total_revenue

from customer_table

group by gender;
```

Result Grid | Filter Rows:

gender	total_revenue
Male	157890
Female	75191

Q.2 which customer used a discount but still spent more than average purchase amount

```
select customer_id, purchase_amount

from customer_table

where discount_applied = 'Yes' and purchase_amount >= (select avg(purchase_amount) from customer_table);
```

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

Q. 3 which are the top 5 highest average review rating

```
select item_purchased ,round(avg(review_rating),2) as average_product_rating

from customer_table

group by item_purchased

order by avg(review_rating) desc

limit 5;
```

item_purchased	average_product_rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

Q.4 compare the average purchase amounts between standard and express shipping

```
select shipping_type,
round(avg(purchase_amount),2)
from customer_table
where shipping_type in ('Standard','Express')
group by shipping_type;
```

shipping_type	average_purchase_amount
Express	60.48
Standard	58.46

Q. 5 do subscribed customer spend more?

```
-- compare average spend and total revenue between subscriber and non-subscriber
select subscription_status,
count(customer_id) as total_customer,
round(avg(purchase_amount),2) as avg_purchase_amount,
sum(purchase_amount) as total_revenue
from customer_table
where subscription_status in ('Yes','No')
group by subscription_status;
```

subscription_status	total_customer	avg_purchase_amount	total_revenue
Yes	10! 1053	59.49	62645
No	2847	59.87	170436

Q.6 top 5 products have the highest percentage of purchase with discounts applied?

```
select item_purchased,
round(sum(case when discount_applied = 'Yes' then 1 else 0 end)/count(*) ,2)  as discount_rate
from customer_table
group by item_purchased
order by discount_rate desc
limit 5;
```

item_purchased	discount_rate
Sneakers	0.50
Hat	0.50
Coat	0.49
Sweater	0.48
Pants	0.47

Q.7 segment customer into new ,returning ,and loyal based on their total

Number of previous purchases and show count of each segment

```
with customer_type as(
select customer_id ,previous_purchases ,
case
   when previous_purchases = 1 then 'New'
```

```
    when previous_purchases between 2 and 10 then 'Returning'
    else 'Loyal'
    end as customer_segment
from customer_table
)
select customer_segment ,count(*) as 'Number of customer'
from customer_type
group by customer_segment;
```

customer_segment	Number of customer
Loyal	3116
Returning	701
New	83

Q.8 what are the top 3 most purchased products within each category

```
with item_count as (
select category ,
item_purchased,
count(customer_id) as total_orders,
Row_Number() over (partition by category order by count(customer_id) desc) as item_rank
from customer_table
group by category ,item_purchased
)
select item_rank,category ,item_purchased,total_orders
from item_count
where item_rank <= 3;
```

Another method :

```
-- select item_rank,category ,item_purchased,total_orders


-- from (
-- select category ,item_purchased,
-- count(customer_id) as total_orders,
-- row_number() over (partition by category order by count(customer_id) desc) as item_rank
-- from customer_table
-- group by category,item_purchased
-- ) as ranked_iem
-- where item_rank <= 3;
```

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163

Q. 9 customer who are repeat buyer (more than 5 previous purchase) also likely to subscribe

```
select subscription_status ,count(customer_id) as repeat_buyers
from customer_table
where previous_purchases >= 5
group by subscription_status;
```

subscription_status	repeat_buyers
Yes	980
No	2583

Q. 10 revenue contribution of each age group

```
select age_group ,
sum(purchase_amount) as total_revenue
from customer_table
group by age_group
order by total_revenue desc;
```

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763